

embeddings:

$$d_l^{\{\text{attn}, \text{mlp}\}} = \frac{1}{|\mathbf{R}|} \sum_{s \in \mathbf{R}} z_l^{\{\text{attn}, \text{mlp}\}}(s) - \frac{1}{|\mathbf{NR}|} \sum_{s \in \mathbf{NR}} z_l^{\{\text{attn}, \text{mlp}\}}(s), \quad (2)$$

where $\mathbf{R}$ and $\mathbf{NR}$ are the sets of reflection and non-reflection steps, respectively.

3.4 Steer model’s reasoning

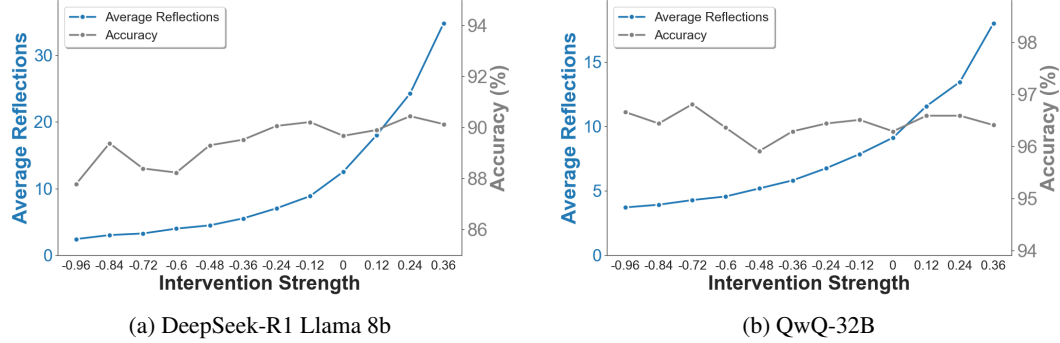


Figure 2: **Accuracy and number of reflection steps under different intervention strengths.** Results are shown for DeepSeek-R1 Llama 8b (distilled model) and QwQ-32B (non-distilled model) on GSM8k. Accuracy remains largely stable, while the number of reflection steps decreases as intervention strength decreases.

With the reflection direction, we can steer the model’s reasoning by injecting this direction into its internal representations. Specifically, the intervention is applied by directly adding the direction vector:

$$z_{l, \text{intv}}^{\{\text{attn}, \text{mlp}\}} = z_l^{\{\text{attn}, \text{mlp}\}} + \lambda d_l^{\{\text{attn}, \text{mlp}\}}. \quad (3)$$

Here, z_l denotes the model output at layer l , and λ is a hyperparameter controlling the intervention strength.

In standard representation-steering approaches, the intervention is applied at **every token generation step**. However, at high intervention strengths, this may push the model’s representation far from the training distribution and degrade model’s performance.

To address this, we propose **stepwise steering**: instead of applying intervention on every token, we apply it only when the model **begins a new thinking step**. Specifically, the intervention is triggered when the last generated token matches the step delimiter “\n\n”. As shown in figs. 4a and 4b, this method preserves intervention effects while avoiding the performance drop observed in full-token steering at high intervention strengths, allowing users to have more control on inference tokens without sacrificing performance.

3.5 Probing model’s uncertainty on reflection direction

Another application of reflection direction is to investigate **RQ1** we proposed in Sec. 1: *When will self-reflection be triggered?* Our hypothesis is that: *Reasoning LLMs trigger reflection when their internal uncertainty is high.*

To verify our hypothesis, we need an approach to quantify the model’s uncertainty during the generation process. Here, we follow [Mielke et al., 2022] and train an auxiliary classifier to predict model answer correctness. For each instance, we compute the projection of the intermediate representation on the reflection direction across all layers. These values are concatenated into a feature vector p_{intv} :

$$p_{\text{intv}} = \text{concat}(\{p_l^{\text{attn}}\}_{l=1}^{N_{\text{layer}}}, \{p_l^{\text{mlp}}\}_{l=1}^{N_{\text{layer}}}), \quad (4)$$

where $p_l^{\{\text{attn}, \text{mlp}\}} = \cos(d_l^{\{\text{attn}, \text{mlp}\}}, z_l^{\{\text{attn}, \text{mlp}\}})$. We extract the feature vector p_{intv} from the end of thinking token (</think> for models we use in the paper), and train a logistic regression model

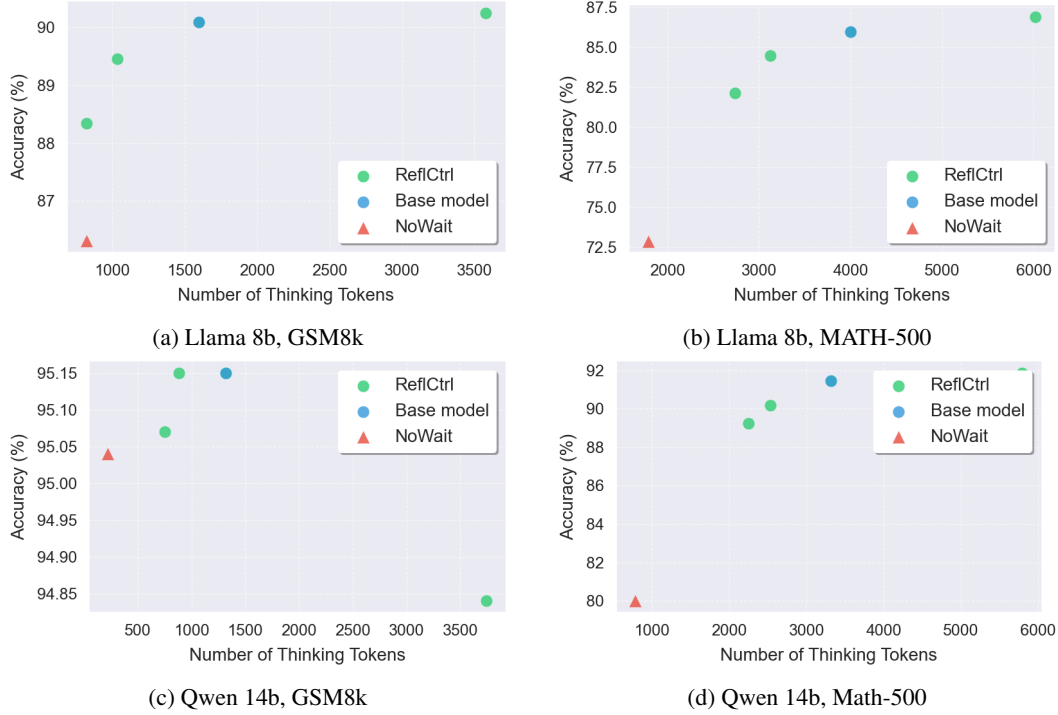


Figure 3: **Accuracy versus reasoning token usage for ReflCtrl compared with NoWait [Wang et al., 2025a].** Results are shown for DeepSeek-R1 Llama 8b and DeepSeek-R1 Qwen 14b across GSM8k and MATH-500 benchmarks. ReflCtrl allows fine-grained control over the trade-off between accuracy and reasoning cost via intervention strength, while NoWait can only suppress reflections entirely. Additionally, ReflCtrl achieves lower performance loss for similar token usage.

upon it on GSM8k training dataset to predict whether model’s answer is correct. If our hypothesis is correct, the classifier should achieve high accuracy, since the reflection direction is aligned with the model’s uncertainty. As the baseline, we use the representation of the last token at the final layer. Results on the GSM8k test set (table 1) show that features derived from the reflection direction achieves higher AUROC and F1 scores, despite having fewer dimensions. This suggests that **model’s uncertainty information is encoded in the reflection direction**, and may be a key factor in triggering self-reflection.

Model	final layer embedding		reflection direction	
	AUROC	F1	AUROC	F1
deepseek-llama-8b	0.736	0.946	0.772	0.948
qwq-32b	0.555	0.636	0.564	0.839
deepseek-qwen-14b	0.716	0.929	0.850	0.976

Table 1: **Probing results for uncertainty detection.** We train a logistic regression classifier to predict answer correctness using (i) the last token embedding at the final layer or (ii) feature vector p_{intv} derived from reflection direction. Reflection-based features achieve higher AUROC and F1 scores despite lower dimensionality, suggesting that uncertainty is encoded in the reflection direction.

4 Experiments

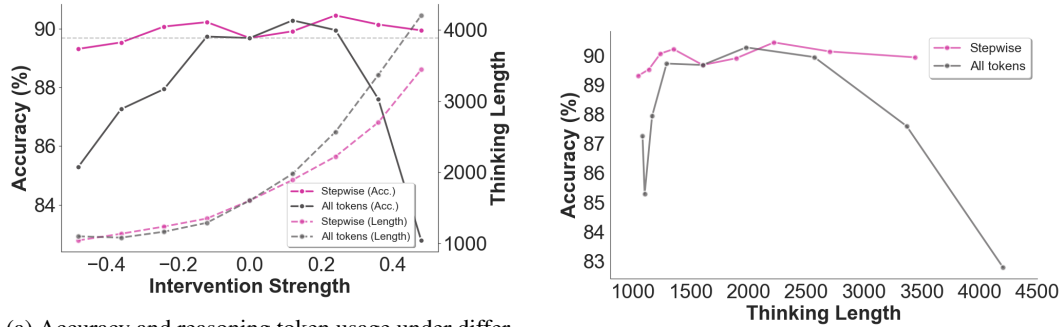
In this section, we conduct an empirical study of our ReflCtrl framework, evaluating how reflection influence model’s performance (**RQ2**) and how our stepwise steering reduces reasoning budget.

4.1 Settings

Models. In this work, we primarily study the DeepSeek-R1-Distilled series of models, including the distilled version of Qwen-2.5 14B and Llama 8B, as these models are publicly available. We also evaluate the QwQ-32B model as a non-distilled reasoning model.

Datasets. For math tasks, we use the GSM8k and MATH-500 as test datasets. For general reasoning tasks, we use the MMLU benchmark, selecting three subsets: Professional accounting, highschool computer science and formal logic.

Generation settings. We follow the standard generation configurations for each model. For math tasks, we use the prompt “Please reason step by step, and put your final answer within \boxed” after each question. For MMLU benchmark, we use the prompt “Please reason step by step, and put your final answer (only the letter) within \boxed.” The maximum completion tokens are set to 8192, except for MATH-500 where we use 16384 due to its higher complexity.



(a) Accuracy and reasoning token usage under different intervention strength.

(b) Accuracy under different reasoning token usage.

Figure 4: Comparison of stepwise versus all-token steering. (a) Accuracy under different intervention strengths when interventions are applied at the start of each reasoning step (stepwise) or at every token (all-token). (b) Accuracy versus reasoning token usage under the two approaches. Stepwise steering preserves accuracy while reducing cost, whereas all-token steering causes significant degradation at large intervention strengths.

Reflection direction extraction. To extract the reflection direction, we use the GSM8k dataset to generate model responses. Then, we apply the method we propose in Sec. 3 to extract the direction. The final step in the reasoning process is omitted as we observe that it is usually a conclusion sentence unrelated to reasoning.

Steering. For the results we show in this section, we apply the stepwise steering method we propose in Sec. 3 unless otherwise specified. The intervention is applied in all layers except the first and last six layers. We further discuss this choice in Sec. 4.4.

4.2 Main experiments

In this section, we study **RQ2: How does reflection influence the model’s reasoning performance.** We begin with DeepSeek-R1 Llama 8b model and test it on GSM8k and MATH-500 datasets with $N = 10$ samples per question. We measure the correctness rate and the reflection rate (number of reflection steps / number of total reasoning steps). As shown in Fig. 5, the correctness rate drops as reflection rate goes higher. However, this does not indicate self-reflection hurts model performance: one possible reason is that the model tends to reflect more when the question is difficult, and the accuracy is generally lower for harder question. We further verify this hypothesis in Sec. C.

To understand the causal relationship between reflection and model performance, we apply different strengths of interventions to the model to intervene self-reflection, and check the model’s performance change under intervention. Fig. 2 shows the accuracy and number of reflection steps under different intervention strengths. We choose DeepSeek-R1 Llama 8b as the distilled model and QwQ-32b for